picture radically by declaring that the “stuff” of all matter and all forces is the same. Each elementary particle is composed of a single string—that is, each particle is a single string—and all strings are absolutely identical. Differences between the particles arise because their respective strings undergo different resonant vibrational patterns. What appear to be different elementary particles are actually different “notes” on a fundamental string. The universe—being composed of an enormous number of these vibrating strings—is akin to a cosmic symphony.

This overview shows how string theory offers a truly wonderful unifying framework. Every particle of matter and every transmitter of force consists of a string whose pattern of vibration is its “fingerprint.” Because every physical event, process, or occurrence in the universe is, at its most elementary level, describable in terms of forces acting between these elementary material constituents, string theory provides the promise of a single, all-inclusive, unified description of the physical universe: a theory of everything (T.O.E.).

The Music of String Theory

Even though string theory does away with the previous concept of structureless elementary particles, old language dies hard, especially when it provides an accurate description of reality down to the most minute of distance scales. Following the common practice of the field we shall therefore continue to refer to “elementary particles,” yet we will always mean “what appear to be elementary particles but are actually tiny pieces of vibrating string.” In the preceding section we proposed that the masses and the force charges of such elementary particles are the result of the way in which their respective strings are vibrating. This leads us to the following realization: If we can work out precisely the allowed resonant vibrational patterns of fundamental strings—the “notes,” so to speak, that they can play—we should be able to explain the observed properties of the elementary particles. For the first time, therefore, string theory sets up a framework for explaining the properties of the particles observed in nature.

At this stage, then, we should “grab hold” of a string and “pluck” it in all sorts of ways to determine the possible resonant patterns of vibration. If string theory is right, we should find that the possible patterns yield exactly the observed properties of the matter and force